

VisionRoot (Pvt) Ltd

Daily Submission Report — Day 09

Date: 29 September 2026

Topic: Advanced Retrieval

- **What I learned:**

- Today I learned about advanced retrieval techniques used in Retrieval-Augmented Generation (RAG) systems.
- I studied the difference between keyword search and semantic search. Keyword search focuses on matching specific words or terms, while semantic search uses embeddings to retrieve information based on meaning.
- I also learned about hybrid search, which combines keyword search and semantic search to improve the relevance of retrieved documents.
- Additionally, I learned about metadata filtering, Top-K retrieval, reranking, and retrieval quality. I understood that the quality of the retrieved context has a major effect on the quality of the final RAG response.
- I also investigated why a RAG system can produce an incorrect answer even when a powerful LLM is used. Incorrect, incomplete, irrelevant, outdated, or excessive retrieved context can lead to an incorrect or unsupported final answer.

What I implemented:

I implemented an advanced retrieval pipeline for a RAG system using Python.

The system includes:

- Keyword search using BM25
- Semantic search using Gemini embeddings
- Hybrid search combining keyword and semantic search
- Metadata filtering using document categories
- Top-K retrieval
- A second-stage reranking process
- Gemini for final answer generation

The implemented flow is:

User Question



Keyword Search + Semantic Search



Hybrid Search



Metadata Filtering



Top-K Retrieval



Reranking



Relevant Context



Gemini



Final Answer

What worked:

- The document collection was successfully stored in ChromaDB with metadata.
- Keyword search successfully retrieved documents based on matching terms.
- Semantic search successfully used Gemini embeddings to retrieve documents based on the meaning of the query.
- Hybrid search successfully combined keyword and semantic retrieval approaches.
- Metadata filtering successfully restricted semantic search results based on document metadata such as category.

- Top-K retrieval allowed me to control the number of documents returned from the search.
- The reranking stage further refined the retrieved candidate documents before they were provided to Gemini.
- The final retrieved context was successfully used to generate answers through the Gemini API.

What failed:

- At the beginning of the task, the sentence-transformers library caused a PyTorch DLL error on Windows:
- WinError 4551: An Application Control policy has blocked this file.
- There was also a ChromaDB persistent storage error caused by the existing chroma_db directory:
- Cannot create a file when that file already exists.
- (os error 183)

How I solved it:

- I first identified that the PyTorch error was caused by a Windows application-control restriction rather than an error in my Python code.
- To avoid the dependency on PyTorch, I changed the semantic embedding implementation from sentence-transformers to the Gemini embedding API.
- I also removed the existing ChromaDB storage directory and recreated the database so that the new collection could be initialized correctly.
- After these changes, the document ingestion and advanced retrieval pipeline worked successfully.

What I would improve:

- I would improve the retrieval system by testing it with a larger and more realistic document collection.
- I would also experiment with different hybrid-search weights and Top-K values to understand their effect on retrieval quality.

- Further improvements could include more advanced reranking models, better document chunking strategies, similarity thresholds, retrieval evaluation metrics, and a larger set of test questions with expected results.

Git commit:

Implement advanced retrieval with hybrid search and reranking

Key Learning:

Keyword Search

= Exact term matching

Semantic Search

= Meaning-based retrieval

Hybrid Search

= Keyword + Semantic Search

Metadata Filtering

= Restrict documents using metadata

Top-K Retrieval

= Control the number of retrieved results

Reranking

= Improve the order of retrieved candidates

Retrieval Quality

= Important factor affecting the final RAG answer

```
(.venv) PS E:\VisionRoot-AI engineer Internship\GitHub\AI-Engineer-Training\AI-Engineer-Training\Week-02\Day-09\practical-task> python ingest.py
Embedding document 9/10...
Embedding document 10/10...
```

```
=====
DOCUMENT INGESTION COMPLETED
=====
Stored documents: 10
Collection: advanced retrieval
Embedding model: gemini-embedding-001
Embedding dimension: 768
=====
```

```
❖ (.venv) PS E:\VisionRoot-AI engineer Internship\GitHub\AI-Engineer-Training\AI-Engineer-Training\Week-02\Day-09\practical-task> |
```

```
(.venv) PS E:\VisionRoot-AI engineer Internship\GitHub\AI-Engineer-Training\AI-Engineer-Training\Week-02\Day-09\practical-task> python app.py
```

```
=====
ADVANCED RETRIEVAL RAG SYSTEM
=====
```

Enter your question.
Type 'exit' to stop.

Question: What is hybrid search?

```
=====
KEYWORD SEARCH RESULTS
=====
```

1. ID: doc4
Category: Retrieval
Document: Hybrid search combines keyword search and semantic search. Keyword retrieval provides exact term matching while semantic retrieval provides meaning-based matching. Combining both methods can improve retrieval quality.
2. ID: doc8
Category: Evaluation
Document: Retrieval quality is affected by document chunking, embedding models, search methods, metadata filters, similarity thresholds, and ranking strategies. Improving retrieval quality can improve the final RAG answer.
3. ID: doc7
Category: Reranking
Document: Reranking is a second-stage retrieval process. An initial retriever produces candidate documents, and a reranker examines the query and candidates more carefully to determine which documents are most

```
=====
SEMANTIC SEARCH RESULTS
=====
```

1. ID: doc4

Category: Retrieval

Document: Hybrid search combines keyword search and semantic search. Keyword retrieval provides exact term matching while semantic retrieval provides meaning-based matching. Combining both methods can improve retrieval quality.

2. ID: doc3

Category: Retrieval

Document: Keyword search looks for exact or closely matching words between the user's query and documents. It works especially well when the user searches for specific technical terms, product names, or identifiers.

3. ID: doc2

Category: Retrieval

Document: Semantic search uses embeddings to understand the meaning of a query. Two sentences can have similar meanings even when they do not contain the same keywords. This makes semantic search useful for natural language questions.

4. ID: doc7

Category: Reranking

Document: Reranking is a second-stage retrieval process. An initial retriever produces candidate documents, and a reranker examines the query and candidates more carefully to determine which documents are most relevant.

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=====
HYBRID SEARCH RESULTS
=====
```

1. ID: doc4

Category: Retrieval

Document: Hybrid search combines keyword search and semantic search. Keyword retrieval provides exact term matching while semantic retrieval provides meaning-based matching. Combining both methods can improve retrieval quality.

Keyword Score: 1.9507

Semantic Score: 0.8749

Hybrid Score: 1.3052

2. ID: doc7

Category: Reranking

Document: Reranking is a second-stage retrieval process. An initial retriever produces candidate documents, and a reranker examines the query and candidates more carefully to determine which documents are most relevant.

Keyword Score: 0.3773

Semantic Score: 0.8076

Hybrid Score: 0.6355

3. ID: doc1

Category: RAG

Document: Retrieval-Augmented Generation, or RAG, combines document retrieval with a language model. The system first retrieves relevant information from a knowledge base and then provides that information to the LLM. RAG can reduce hallucinations when the retrieved information is relevant and accurate.

Keyword Score: 0.3213

```

3. ID: doc3
Category: Retrieval
Document: Keyword search looks for exact or closely matching words between the
         user's query and documents. It works especially well when the user
         searches for specific technical terms, product names, or identifiers.
Keyword Score: 0.0000
Semantic Score: 0.8104
Hybrid Score: 0.4862
Rerank Score: -5.9312
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.

=====
FINAL ANSWER
=====
Based on the provided context, hybrid search is a method that combines keyword search and semantic search. Keyword retrieval provides exact term matching, while semantic retrieval provides meaning-based matching.

Question: 

```

Question: How can a system understand the meaning of a question instead of only matching exact words?

```

=====
KEYWORD SEARCH RESULTS
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1. ID: doc2
Category: Retrieval
Document: Semantic search uses embeddings to understand the meaning of a query.
         Two sentences can have similar meanings even when they do not contain
         the same keywords. This makes semantic search useful for natural
         language questions.

2. ID: doc3
Category: Retrieval
Document: Keyword search looks for exact or closely matching words between the
         user's query and documents. It works especially well when the user
         searches for specific technical terms, product names, or identifiers.

3. ID: doc4
Category: Retrieval
Document: Hybrid search combines keyword search and semantic search. Keyword
         retrieval provides exact term matching while semantic retrieval
         provides meaning-based matching. Combining both methods can improve
         retrieval quality.

4. ID: doc1
Category: RAG
Document: Retrieval-Augmented Generation, or RAG, combines document retrieval
         with a language model. The system first retrieves relevant information

```